

MOHAND OUIDIR AMIRAT

3RD YEAR PhD CANDIDATE AT UNIVERSITÉ CLAUDE BERNARD LYON 1

BORN ON JULY 02, 1998 IN ANNABA, ALGERIA / NATIONALITY: ALGERIAN

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RESEARCH INTERESTS

My research interests lie within the field of Control Theory for Partial Differential Equations (PDEs). I am particularly interested in the analysis and stabilization of nonlinear hyperbolic equations, with a specific focus on *weakly hyperbolic systems of conservation laws*. My work also involves numerical analysis and scientific computing, essentially for models arising from process engineering and fluid mechanics.

EDUCATION

PhD Candidate in Automatic Control (Specialization: Control Theory) <i>Université Claude Bernard Lyon 1 (LAGEPP) - Lyon, France</i>	<i>2023 - Pres.</i>
Thesis Title: Boundary stabilization of 1-D weakly hyperbolic systems of conservation laws: application to sedimentation in wastewater treatment	
Supervisors: Vincent Andrieu , Mathieu Bajodek and Claire Valentin	
Funding: Doctoral fellowship from the French Ministry of Research (MESRI) obtained via the EEA Doctoral School (Lyon)	
M.Sc. in Mathematics - Modeling and Numerical Analysis (Full 2-year program) <i>Université de Montpellier - Montpellier, France</i>	<i>2021 - 2023</i>
1st year M.Sc. in Applied Mathematics / Rank: 2 nd in class <i>Université Badji Mokhtar - Annaba, Algeria</i>	<i>2020 - 2021</i>
B.Sc. in Mathematics / Rank: 3 rd in class <i>Université Badji Mokhtar - Annaba, Algeria</i>	<i>2017 - 2020</i>
High School Diploma / Distinction: Honors (Assez Bien) <i>Specialization: Technical-Mathematics - Algeria</i>	<i>2016 - 2017</i>

PUBLICATIONS

Articles in International Journals

- **M.O. Amirat**, M. Bajodek, V. Andrieu, J. Auriol, C. Valentin. *Insights into the stability analysis of 2-by-2 linear weakly hyperbolic systems*. Accepted for publication in *Automatica*, 2026.
- J. Auriol, A. Braun, **M. O. Amirat**, M. Bajodek. Underactuated boundary control of a linearized Saint-Venant equation. *IMA Journal of Mathematical Control and Information*, 2025.
- *In preparation* – **M.O. Amirat**, V. Andrieu, M. Bajodek, C. Valentin. *Stabilization of a shock steady-state for weakly hyperbolic systems of balance laws: application to wastewater treatment*.

Conference Proceedings

- **M.O. Amirat**, V. Andrieu, J. Auriol, M. Bajodek, C. Valentin. *Boundary feedback control of a 2×2 weakly hyperbolic system: Lyapunov-based approach*. *IFAC-PapersOnLine*, Vol. 59, Issue 8, 2025
- **M.O. Amirat**, V. Andrieu, M. Bajodek, J. Auriol, C. Valentin. *Stability analysis of a class of 2×2 triangular hyperbolic systems*. *IFAC-PapersOnLine*, Vol. 58, Issue 21, 2024.

TALKS & COMMUNICATIONS

International Conferences

- 5th IFAC/IEEE-CSS Workshop on Control of Systems Governed by PDEs (CPDE 2025), Beijing, China *June 2025*
- 4th IFAC Conference on Modelling, Identification and Control of Nonlinear Systems (MICNON 2024), Lyon, France *Sept. 2024*

National Conferences

- Congrès des Jeunes Chercheur.e.s en Mathématiques Appliquées (CJC-MA 2026), ENPC, Champs-sur-Marne, France. March 2026
- 2^{ème} Congrès annuel de la société d'Automatique, de Génie Industriel et de Productique (SAGIP). Villeurbanne, France. May 2024

Seminars & Workshops

- Spring School on Theory and Applications of Port-Hamiltonian Systems (PHS 2025), Fraueninsel, Germany. March 2025
- DYCOP Team Seminar, LAGEPP. Université Claude Bernard Lyon 1, France. (2024, 2025 and 2026)

TEACHING EXPERIENCE

Teaching assistant at the Department of Electrical and Process Engineering *2023 – Present*
Université Claude Bernard Lyon 1

Total: 144 teaching hours

Numerical Approximation of PDEs (2nd year graduate students in Control Engineering)

Design and teaching of the last part of the course "Distributed Parameter Systems" (numerical part).

- Finite Difference schemes for Heat and Advection-Diffusion equations.
- Finite Volume methods for nonlinear hyperbolic systems (1-D Saint-Venant).
- Godunov schemes and approximate Riemann solvers, implementation in Matlab.

Numerical Methods for Engineers (1st year Graduate students in Electrical Engineering)

Sparse linear systems (direct/iterative methods) and numerical optimization.

Advanced Programming: C Language (3rd year undergraduate students in Electrical Engineering)

Pointers and arrays, memory allocation, modular programming (sources, headers, makefiles), and low-level programming (bitwise operators, masks, bit-fields).

Introduction to Automatic Control (2nd year undergraduate students in Electrical Engineering)

Laplace transform and Convolution, stability, frequency domain and elementary feedback laws (P, PI, PID).

RESPONSIBILITIES

Co-organizer of the PhD Seminar (Café Contrôle)

2025 - Pres.

LAGEPP, Université Claude Bernard Lyon 1 - Lyon, France

Mission: Speaker selection, logistical management, and session moderation to foster scientific exchange among PhD students and Post-Doc researchers in the laboratory.

RESEARCH EXPERIENCE & MASTER INTERNSHIPS

2nd year M.Sc. Internship: Development of a Godunov-type Finite Volume scheme for the numerical approximation of a weakly hyperbolic system

March – Aug 2023

Application to sedimentation in wastewater treatment

LAGEPP, Université Claude Bernard Lyon 1 - Lyon, France

Supervision: Frédéric Lagoutière, Claire Valentin and Christian Jallut

Abstract: Implementation of a Finite Volume solver for nonlinear hyperbolic systems. Development of splitting strategies. Comparison with experimental data and parameter estimation through optimization.

1st year M.Sc. Internship: Numerical methods for Hamilton-Jacobi equations and viscosity solutions

March – June 2022

IMAG, Université de Montpellier - Montpellier, France

Supervision: Jessica Guerand

Abstract: Theoretical and numerical study of Hamilton-Jacobi equations and viscosity solutions. Extension to the non-stationary case, including the analysis and implementation of a Finite Difference scheme with optimal error estimates.

TECHNICAL SKILLS

Languages: C/C++, MATLAB, Python, FreeFEM++, FEniCS, HTML/CSS

Software & Tools: ParaView, Gnuplot, Gmsh, L^AT_EX, Git/GitHub, VS Code

Operating Systems: macOS, Linux (Ubuntu), Windows

LANGUAGES

French: Native/Bilingual (C2) | **English:** Advanced (C1)

Berber: Mother language | **Arabic:** Native/Bilingual (C2)